1 November 2025, 제주

한 국 우 주 과 학 회　가 을 학 술 대 회

Prediction of Heavy Ion Charge Sate Ratios & Elemental Composition of Solar Wind Using Deep Learning

서정준[1], 손지현[1], 이현수[2], 김지영[2], & 이경선[3]

[1] 한국천문연구원, 태양권 연구센터
[2] 충남대학교, 천문우주과학과
[3] 서울대학교, 물리천문학부

발표 내용

1. 서론

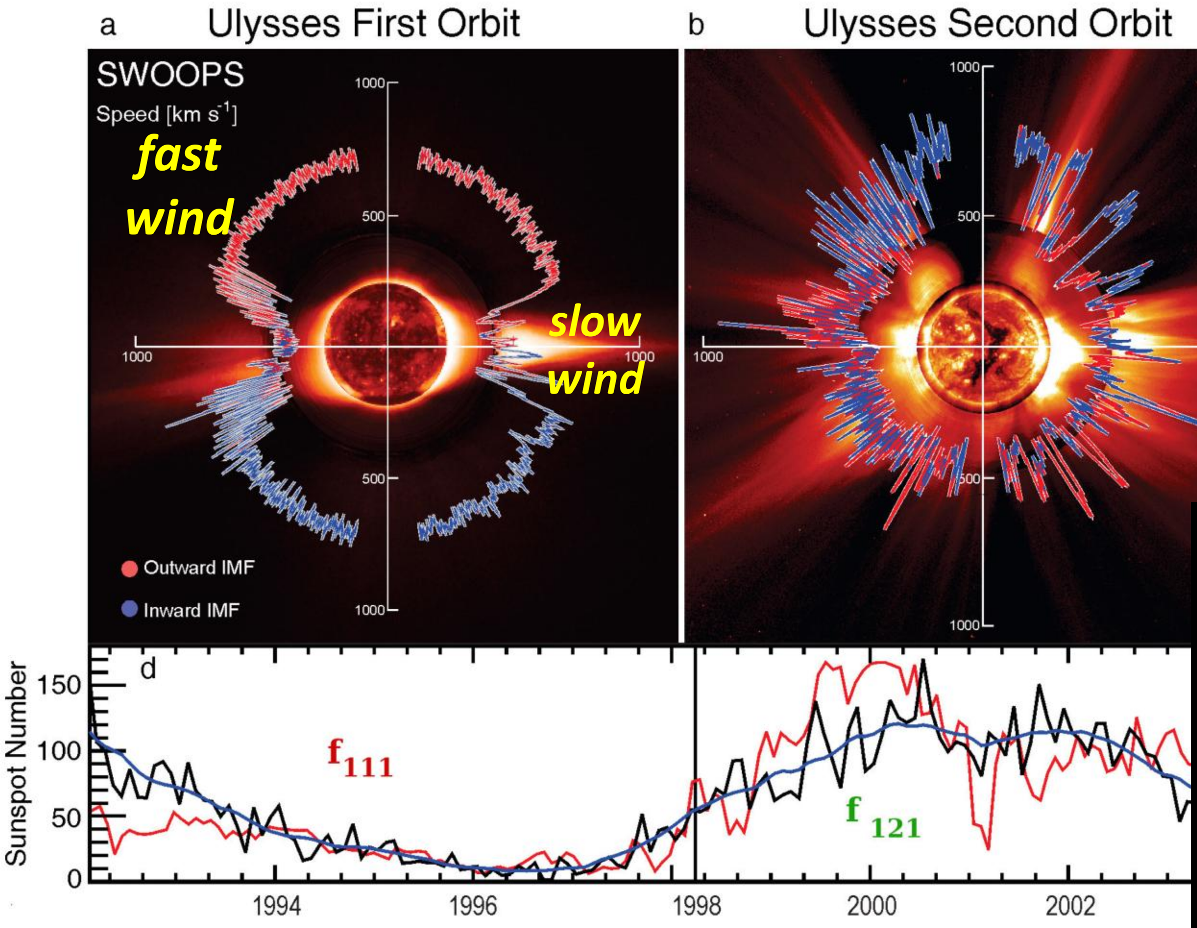
- ✓ 태양풍 근원
- ✓ 중이온 이온화 비율
- ✓ 연구의 의미 및 중요성

2. 태양풍 중이온 이온화 비율 예측 모델

- ✓ ACE 위성 데이터 1998 - 2011
- ✓ input paramters 선정 요인
- ✓ 중이온 이온화 비율 예측 모델 결과

3. 결론 및 논의

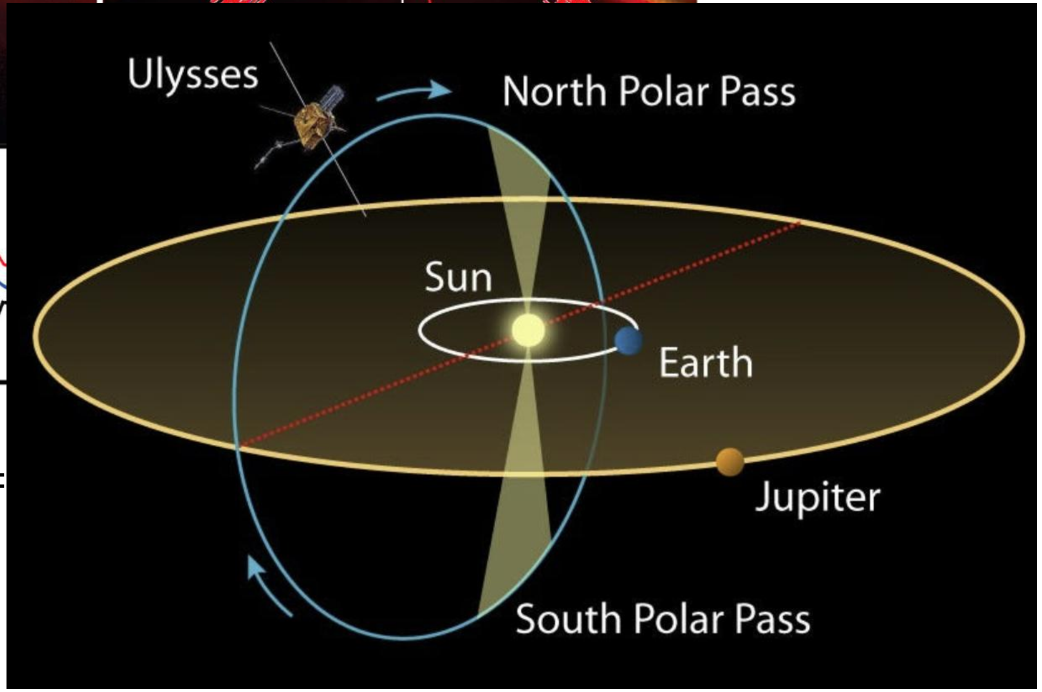
Solar wind by Ulyssess mission



흑점 개수

McComas et al., 2008, GRL

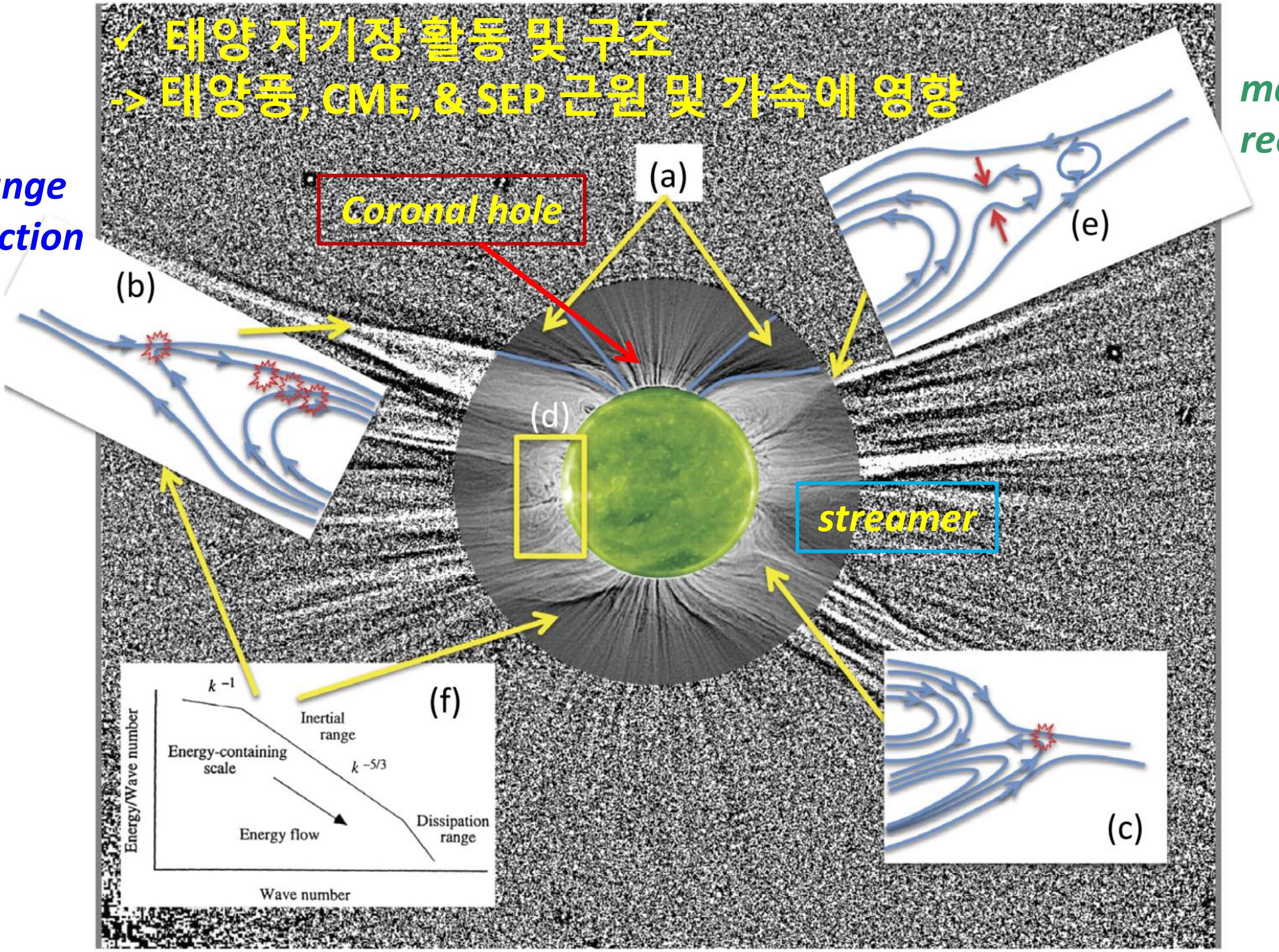
1990 ~ 2009년 태양



✓ ***Solar wind structure***
depends mainly on the solar magnetic activity.

Possible sources of slow solar wind

interchange reconnection

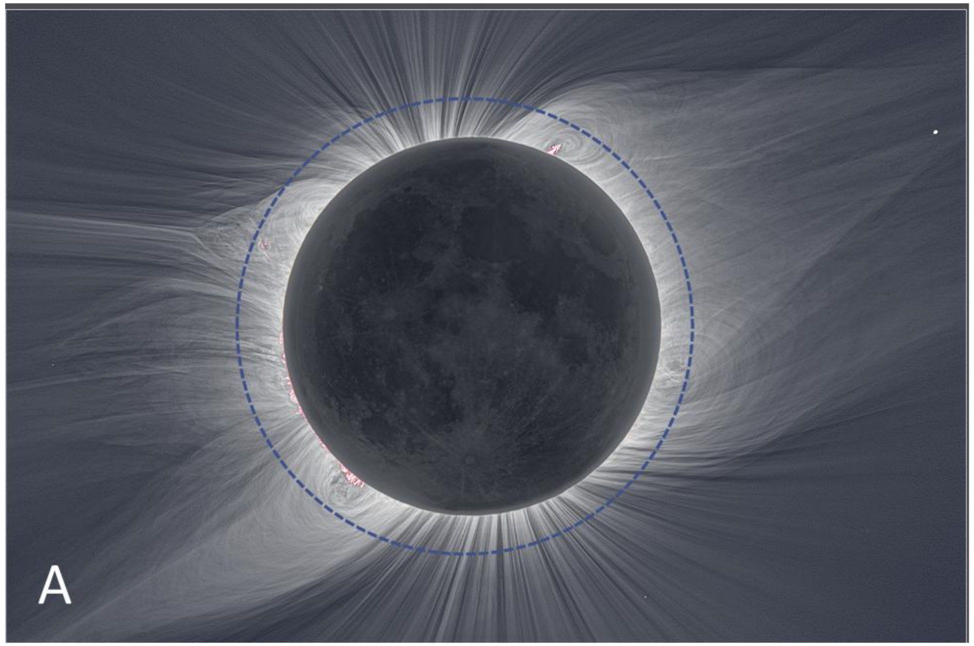


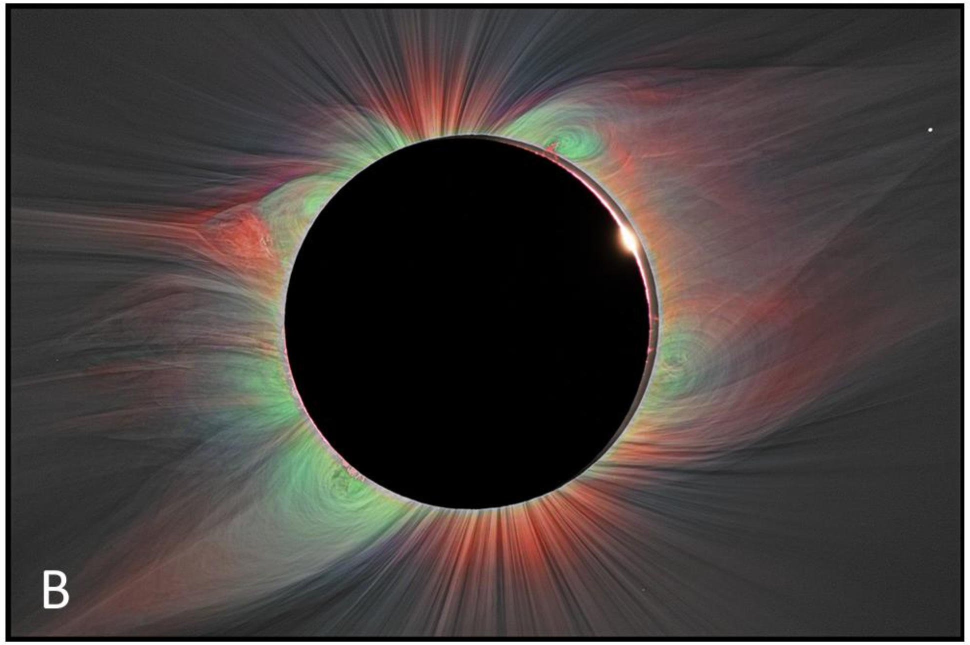
magnetic reconnection

near the boundary layer
: Alfven-wave turbulence

Abbo et al., 2017, SP

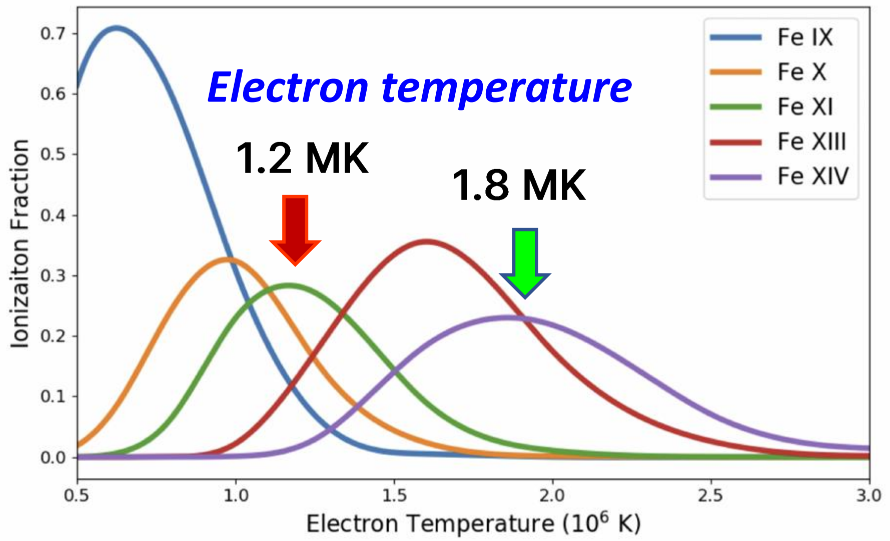
Total Solar Eclipse 1 Aug 2008





Habbal et al., 2010, ApJ

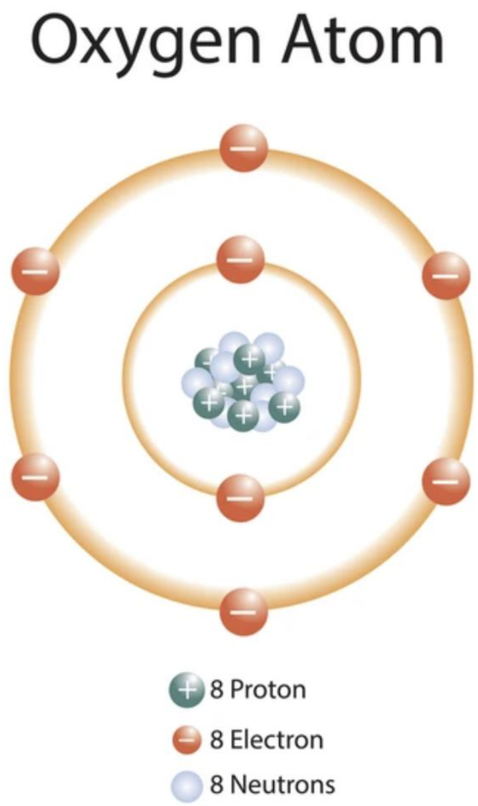
C

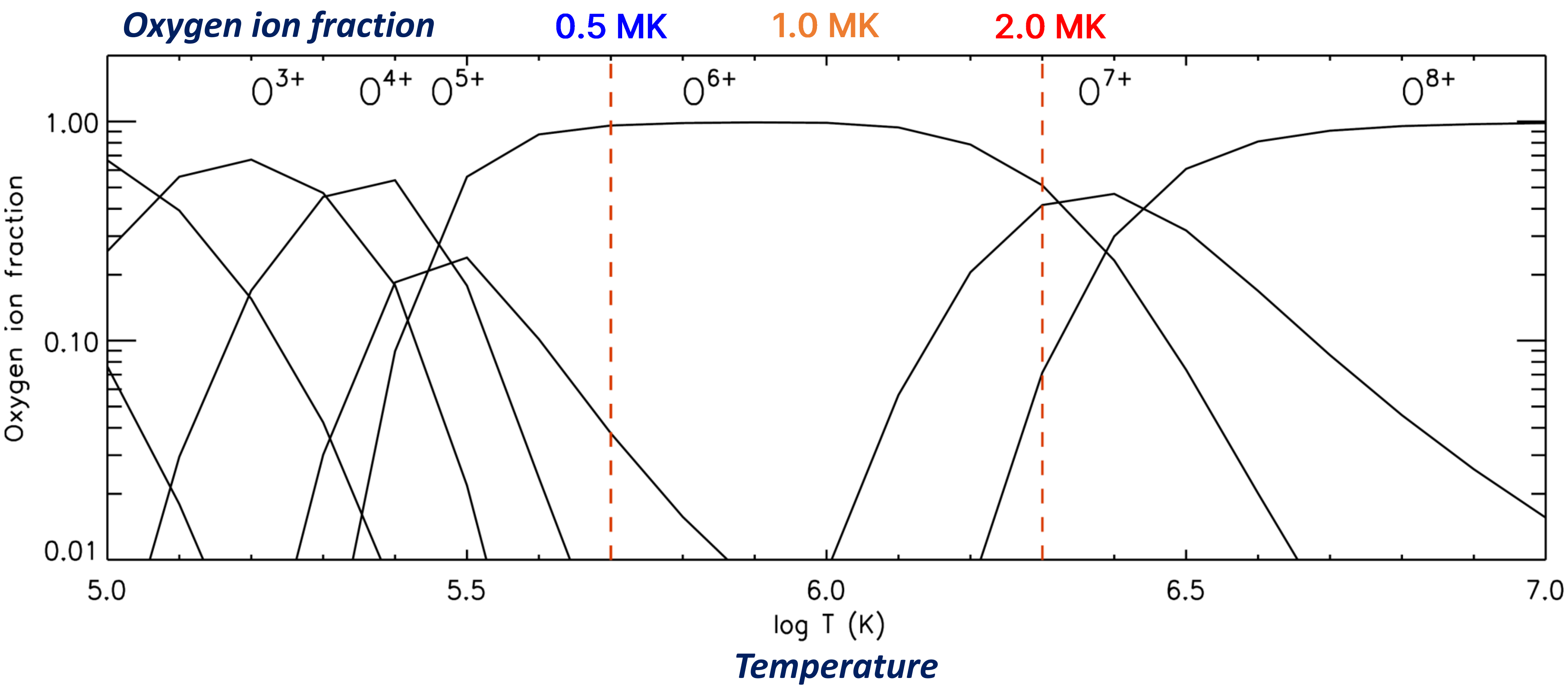


✓ Red: Fe X & Fe XI - **1.2 MK**

✓ Green: Fex III & Fe XIV - **1.8 MK**

Oxygen ion abundance
as a function of temperature under ionization equilibrium



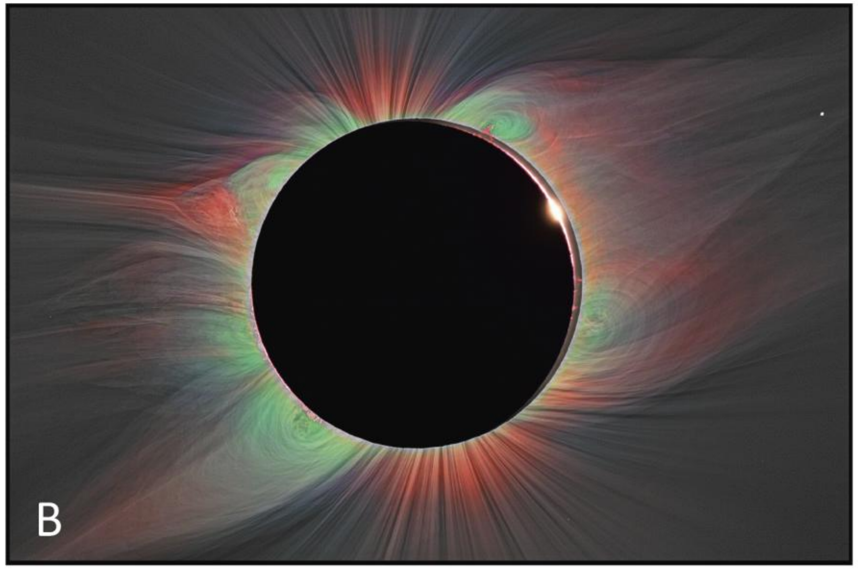


- ✓ **Coronal temperature:** The Oxygen ionization state is "frozen-in" at a certain temperature as it expands from the corona.
- ✓ Higher coronal temperatures lead to higher charge states in the solar wind.

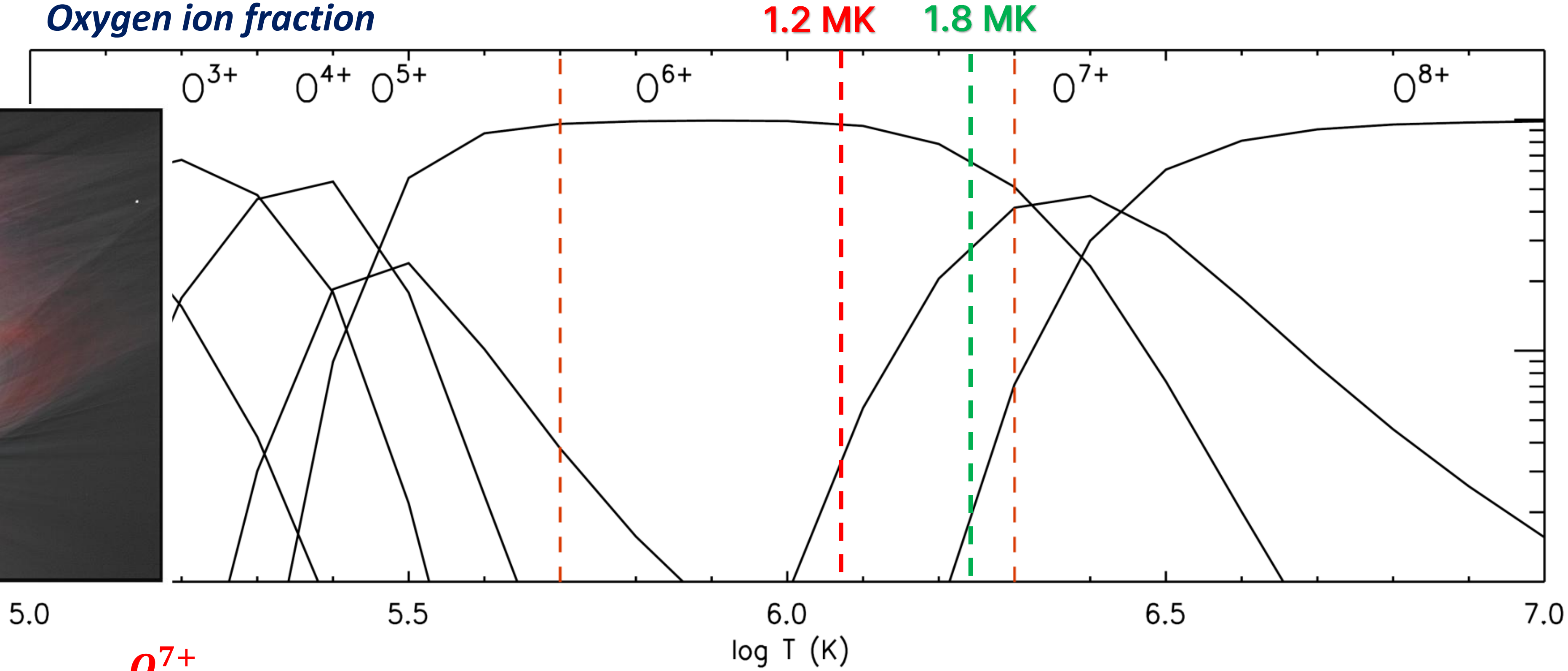
Oxygen ion abundance

as a function of temperature under ionization equilibrium

Total Solar Eclipse 1 Aug 2008



Oxygen ion fraction



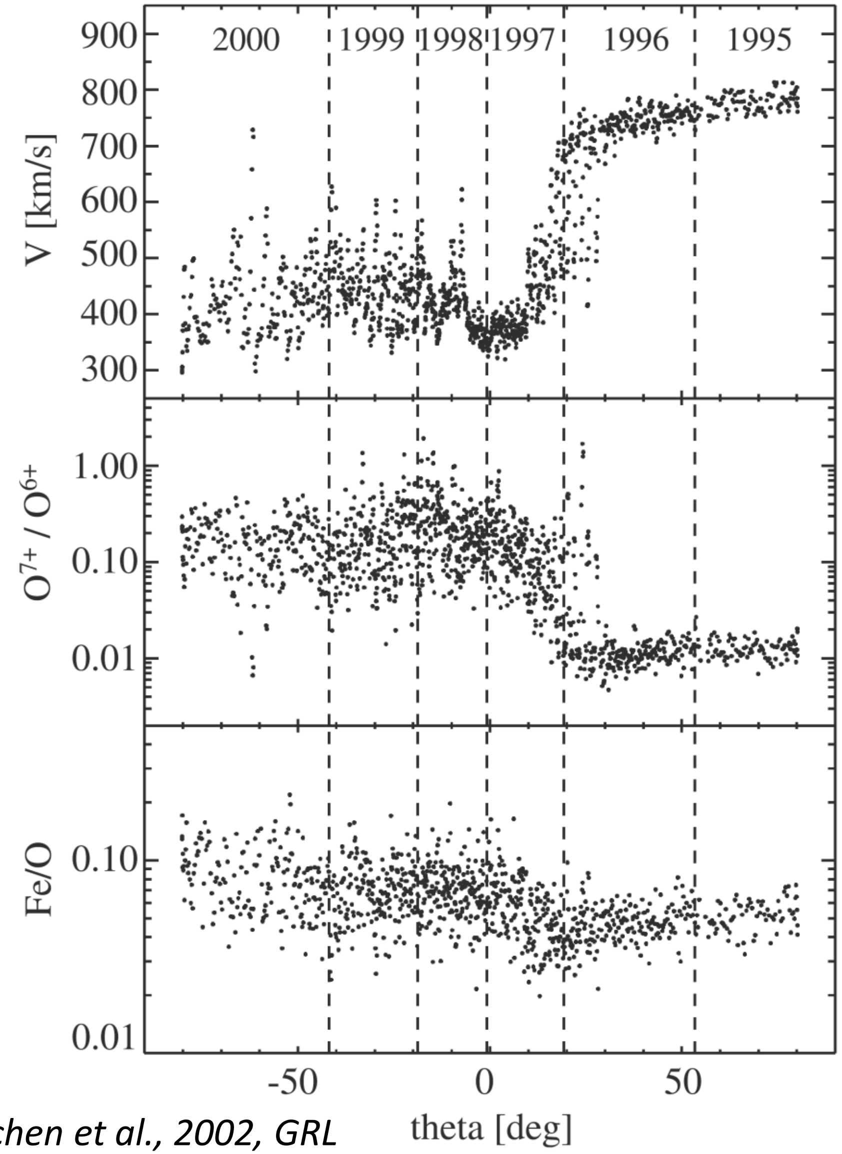
Temperature

✓ Red: Fe X & Fe XI - **1.2 MK** $\frac{O^{7+}}{O^{6+}} \approx 0.05$

✓ Green: Fex III & Fe XIV - **1.8 MK** $\frac{O^{7+}}{O^{6+}} \approx 0.5$

✓ ***It is expected that O^{7+}/O^{6+} are inversely correlated with wind speed.***

Ulysses observation

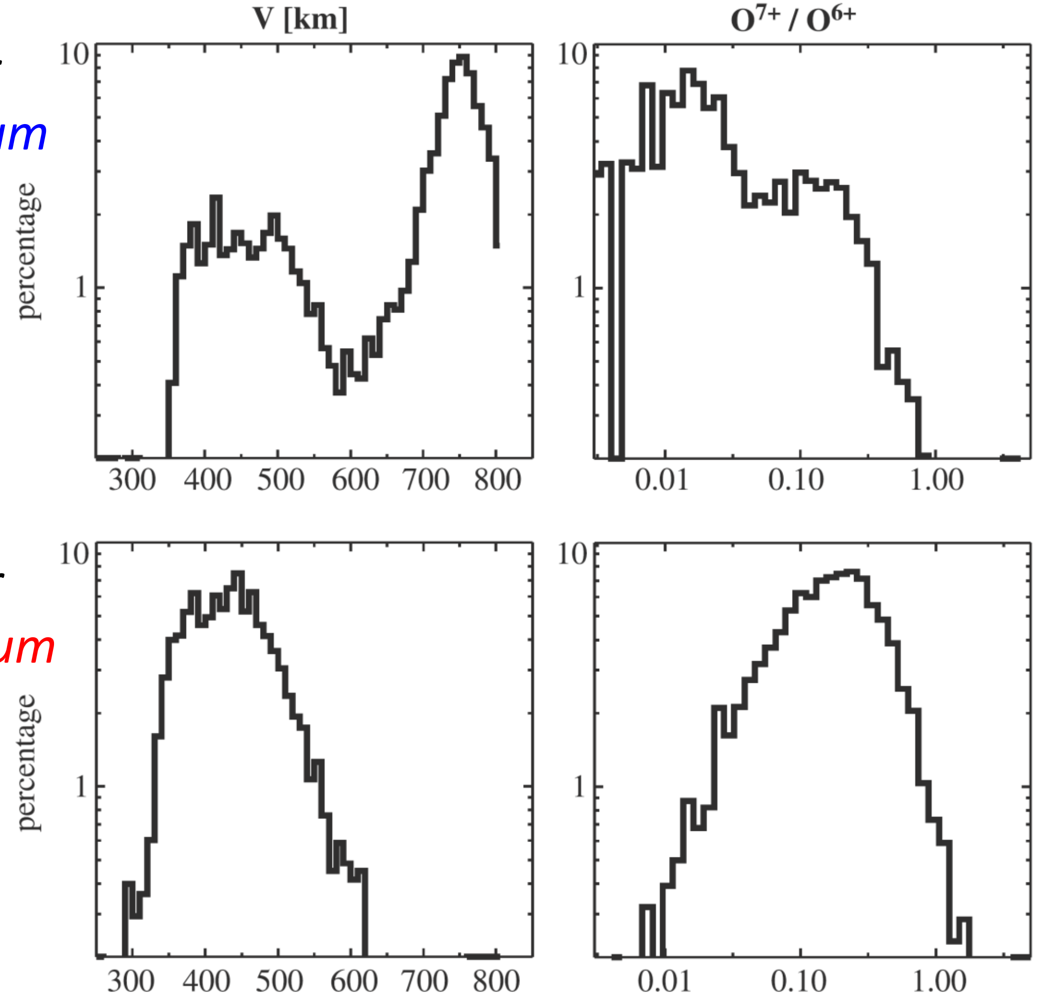


Zurbuchen et al., 2002, GRL

SW speed ***O^{7+}/O^{6+}***

Solar minimum

Solar maximum

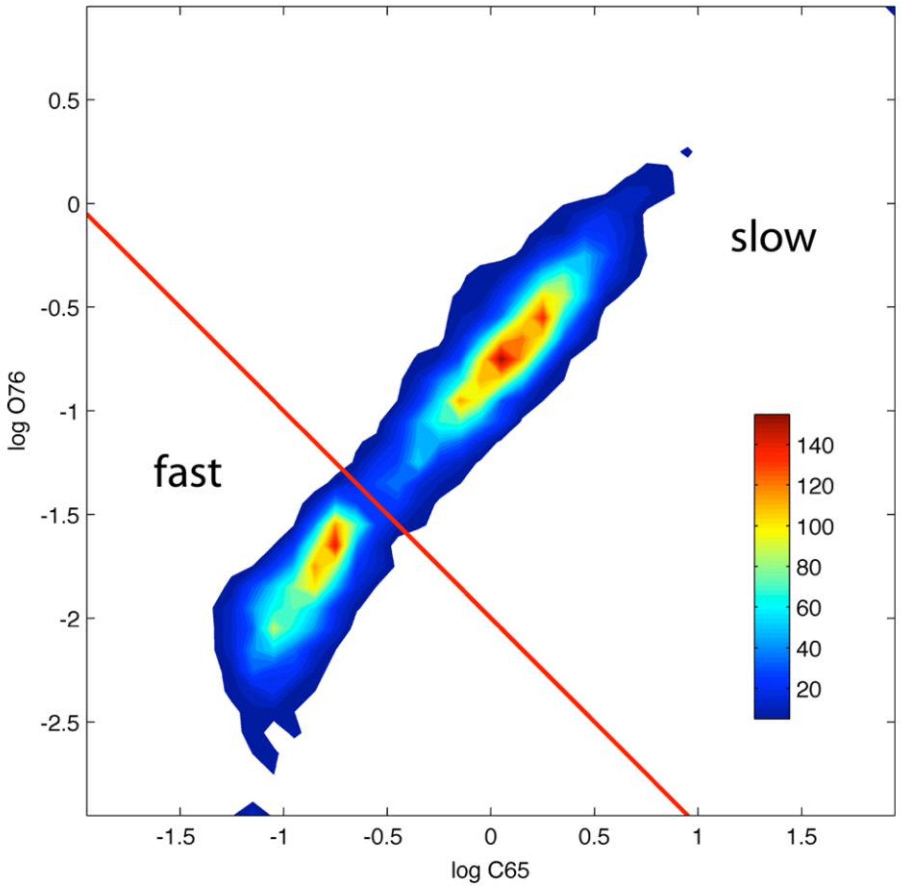


✓ **Fast solar wind**, originating from coronal holes, has consistently lower charge states.

Ulysses observation

Oxygen ratio (O^{7+}/O^{6+}) *and Carbon ratio* (C^{6+}/C^{5+})

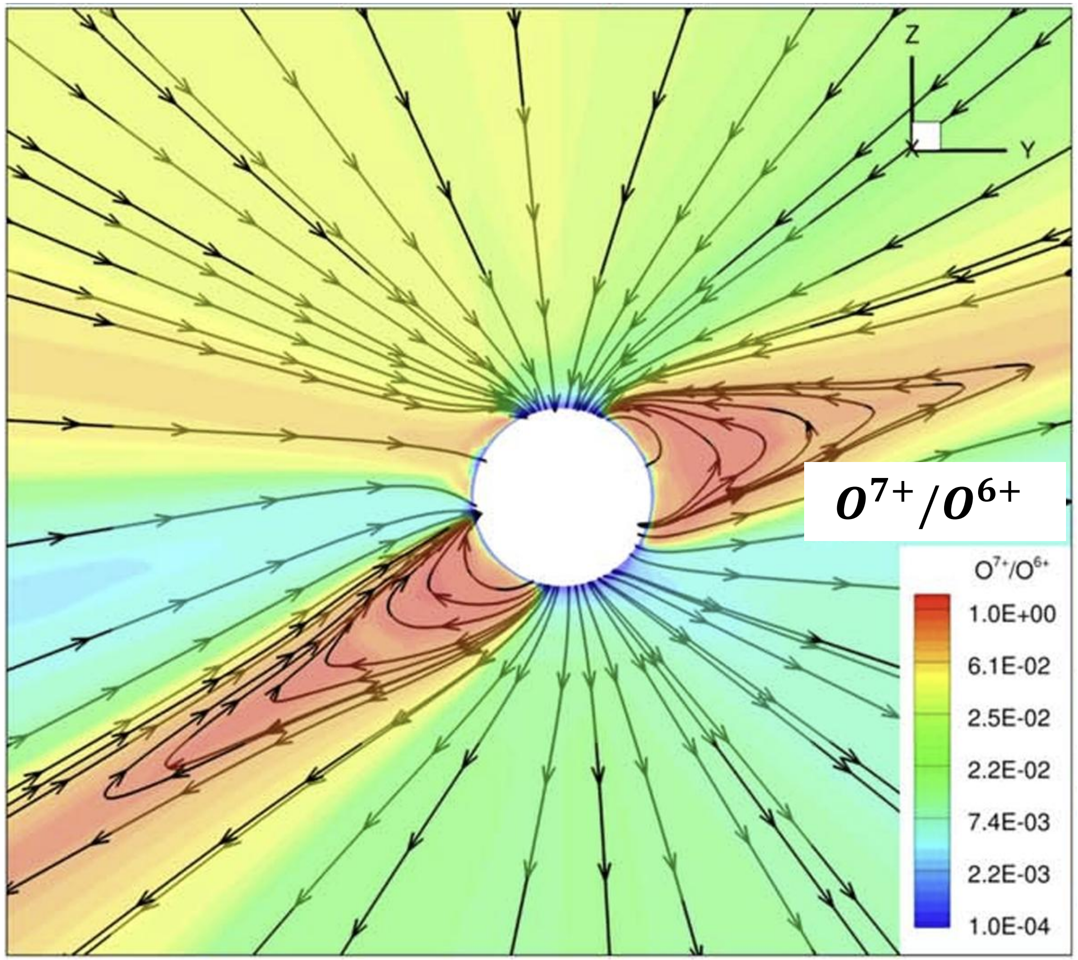
$log\ O^{7+}/O^{6+}$



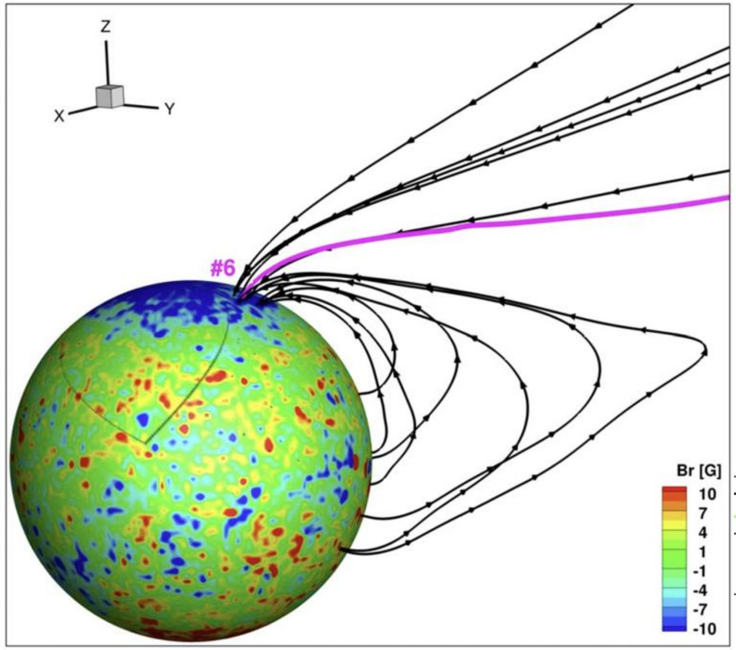
$log\ C^{6+}/C^{5+}$

Abbo et al., 2016, SSR

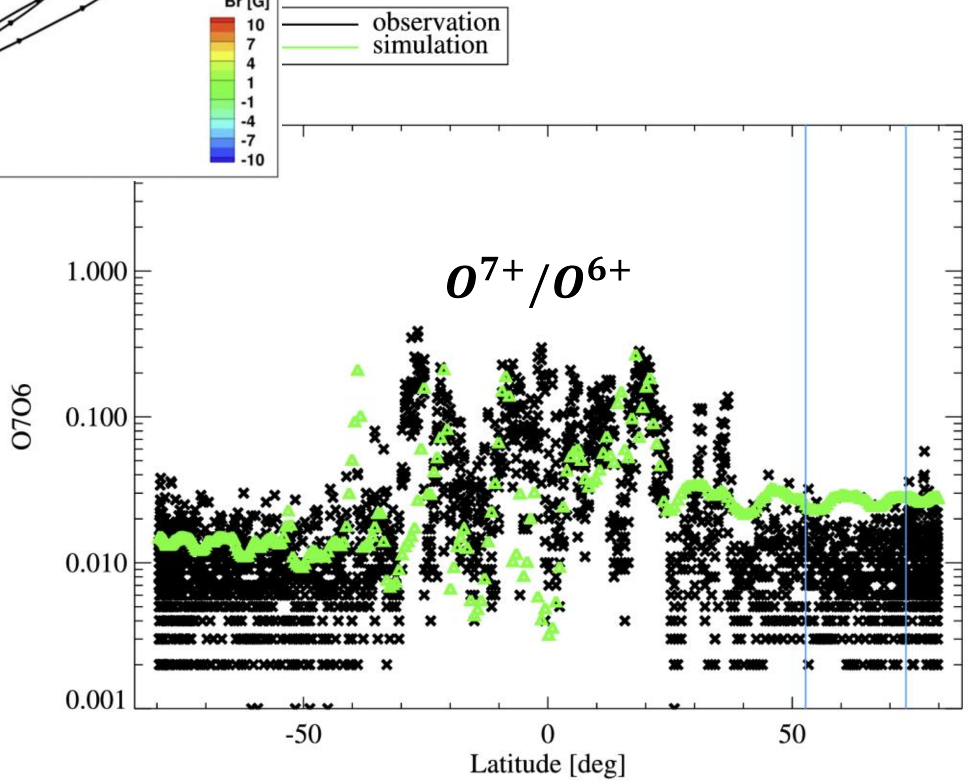
연구 의미 및 중요성 1 : 태양풍 근원



Charge state calculation
for global solar wind modeling

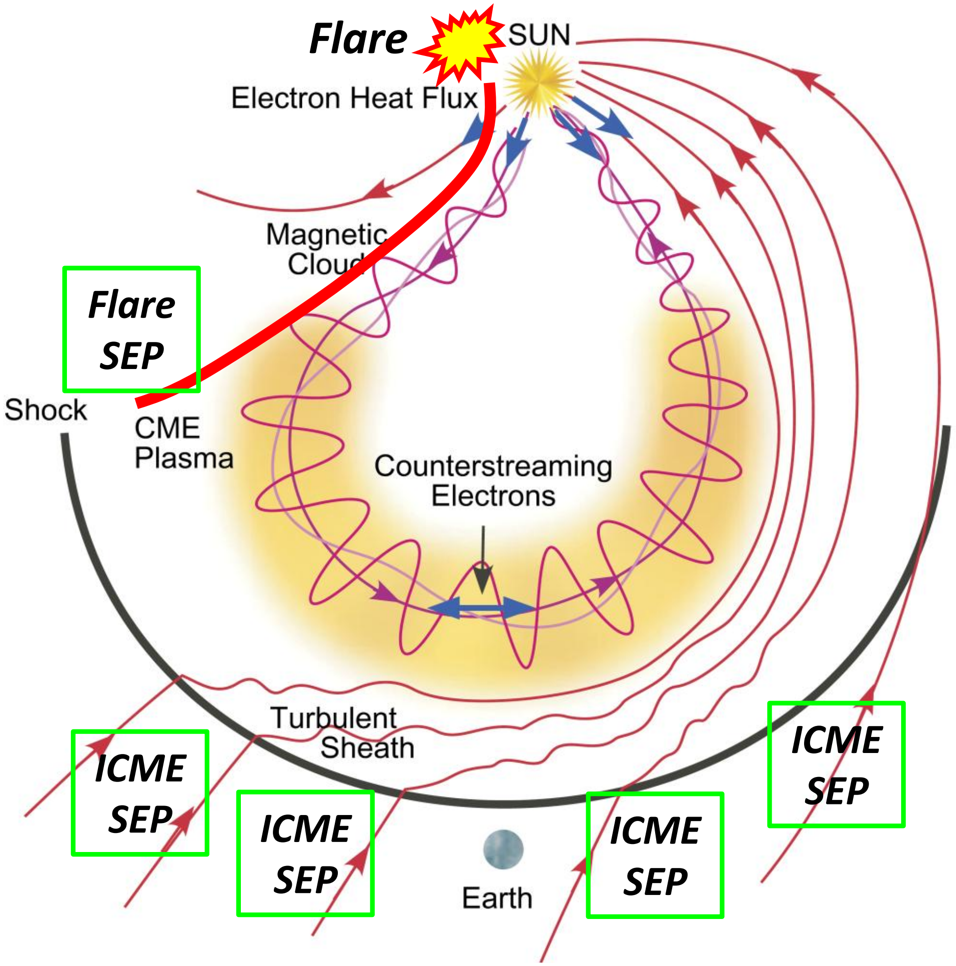


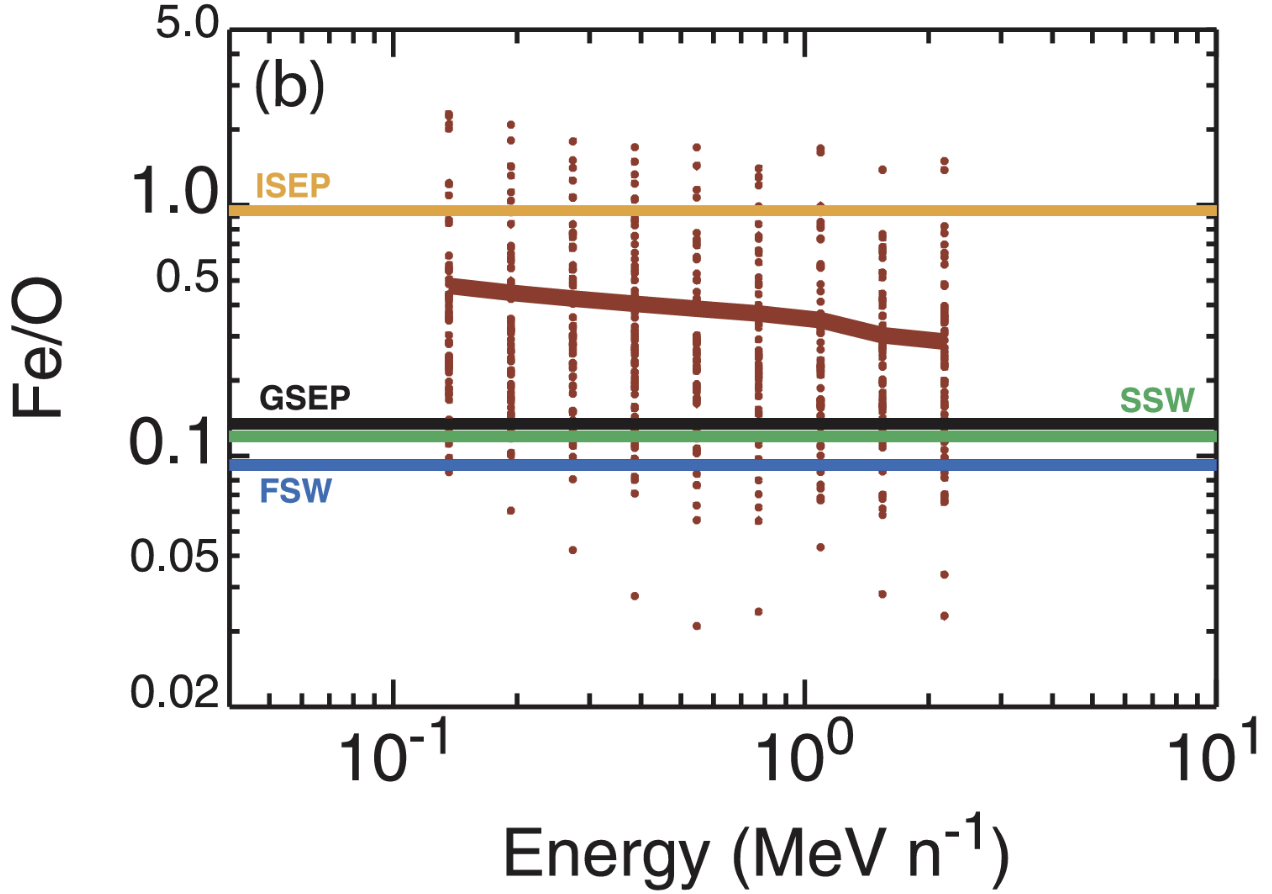
Szente et al., 2022, ApJ



연구 의미 및 중요성 1 : 태양풍 근원

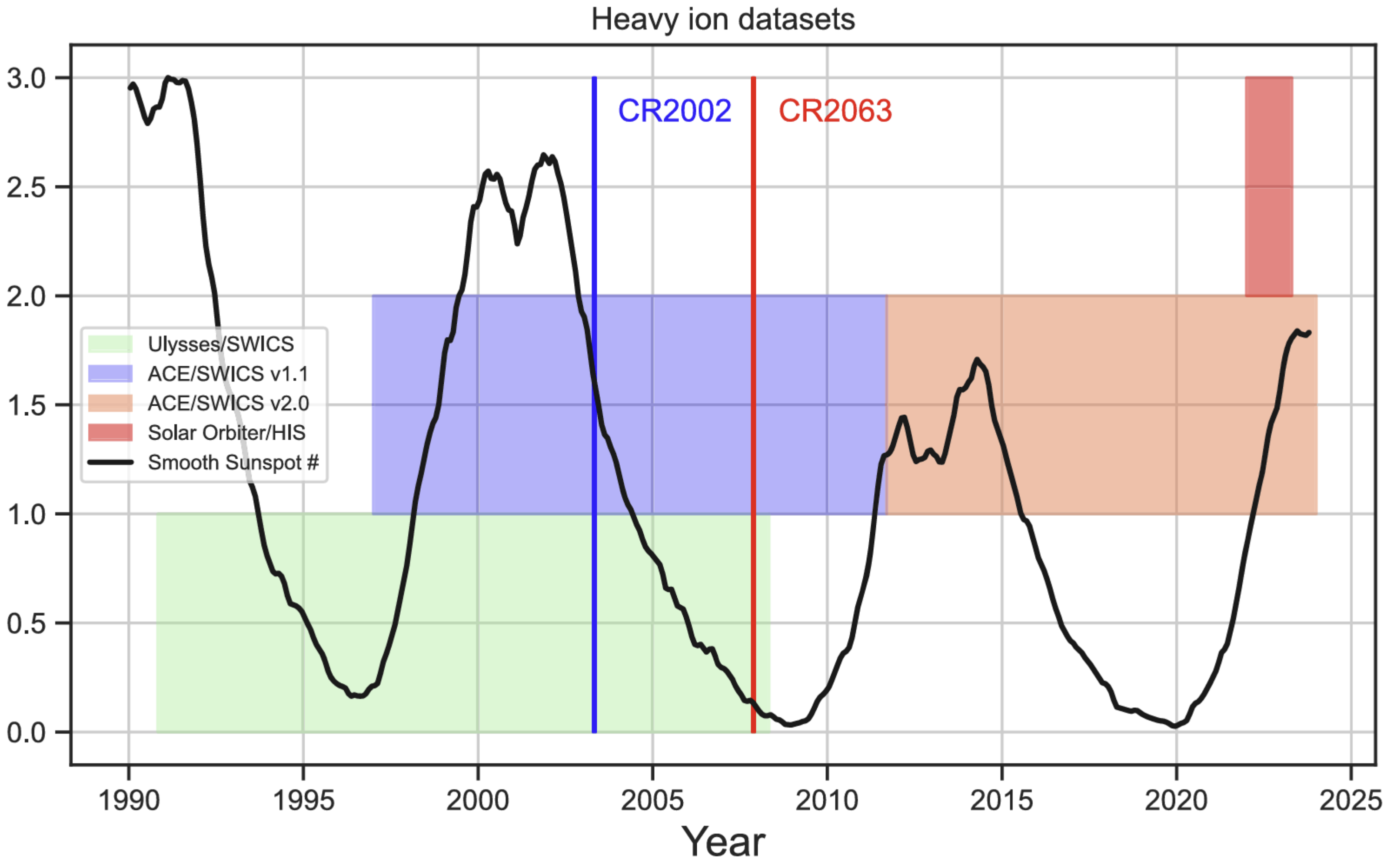
Standard picture of "ICME"





연구 의미 및 중요성 2: 중이온 현장 관측기

Timeline exhibiting for availability of heavy ion in situ measurements



Data set for deep learning

1. **ACE spacecraft :** 태양풍 양성자 및 자기장 자료

✓ **Input** data (64초): Solar wind speed, density, temperature, helium abundance,
proton specific entropy, Coulomb number, & Alfvenicity (SWEPAM & MAG)

✓ **Target** data (2시간): $\boldsymbol{O^{7+}/O^{6+}}$, $\boldsymbol{C^{6+}/C^{5+}}$, **&** $\boldsymbol{Fe/O}$ **-** 중이온 이온화 비율 (SWICS)

2. **Period:** Feb 1998 – Aug 2011, 태양 극대/극소기 포함 약 13년

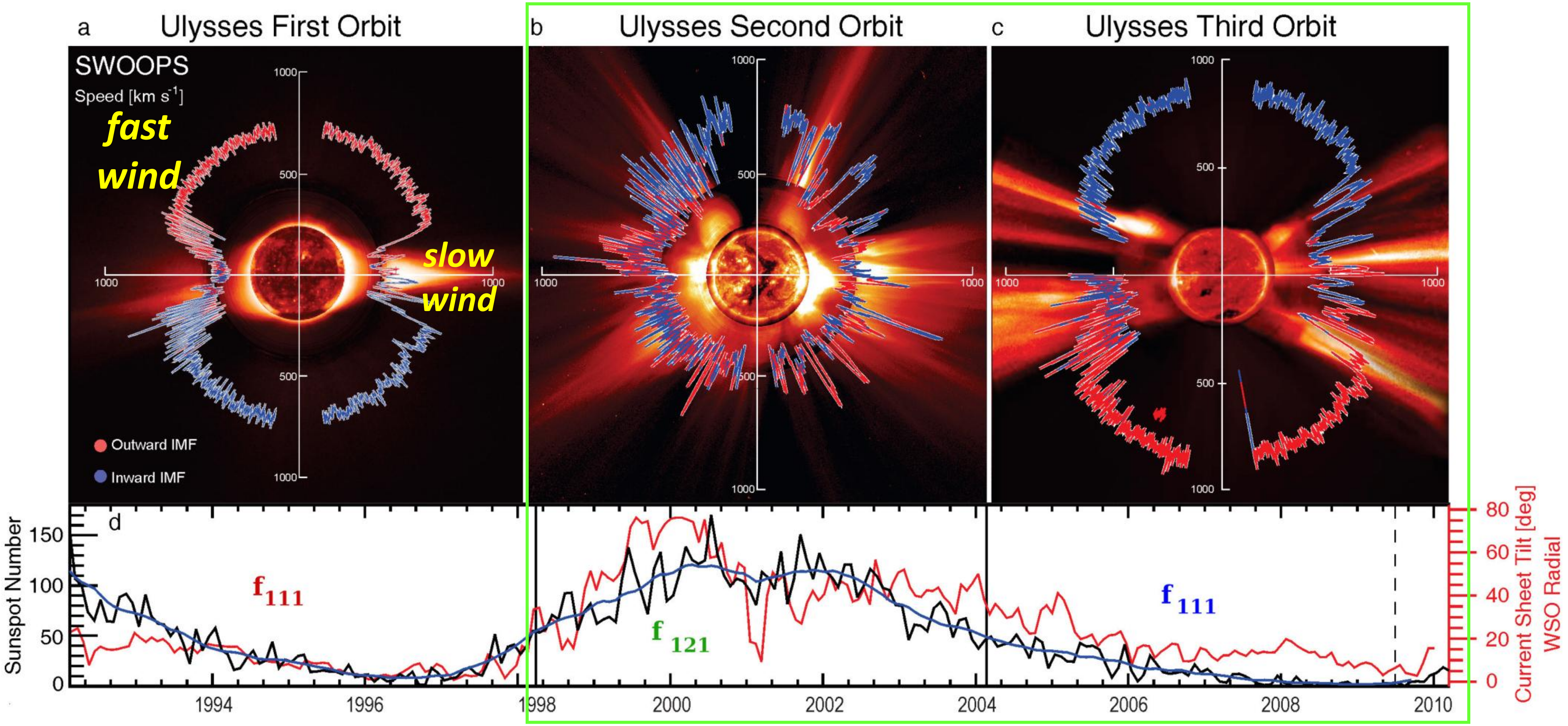
✓ Training set: 매년 1 – 8월

✓ Validation set: 매년 9월

✓ Test set: 매년 10 – 12월

✓ Target data 개수: 12개(1일) x 365일(1년) x 13년 ~ 약 50,000개

Data set for deep learning



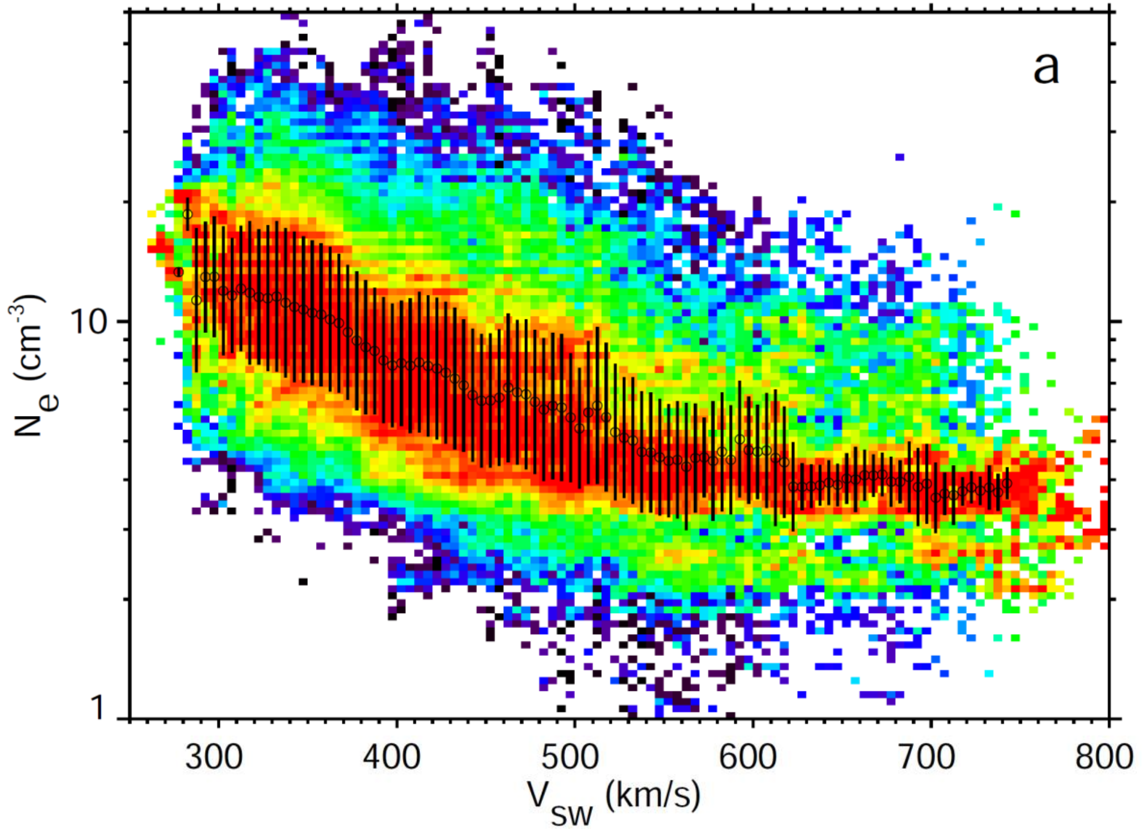
흑점 개수

McComas et al., 2008, GRL

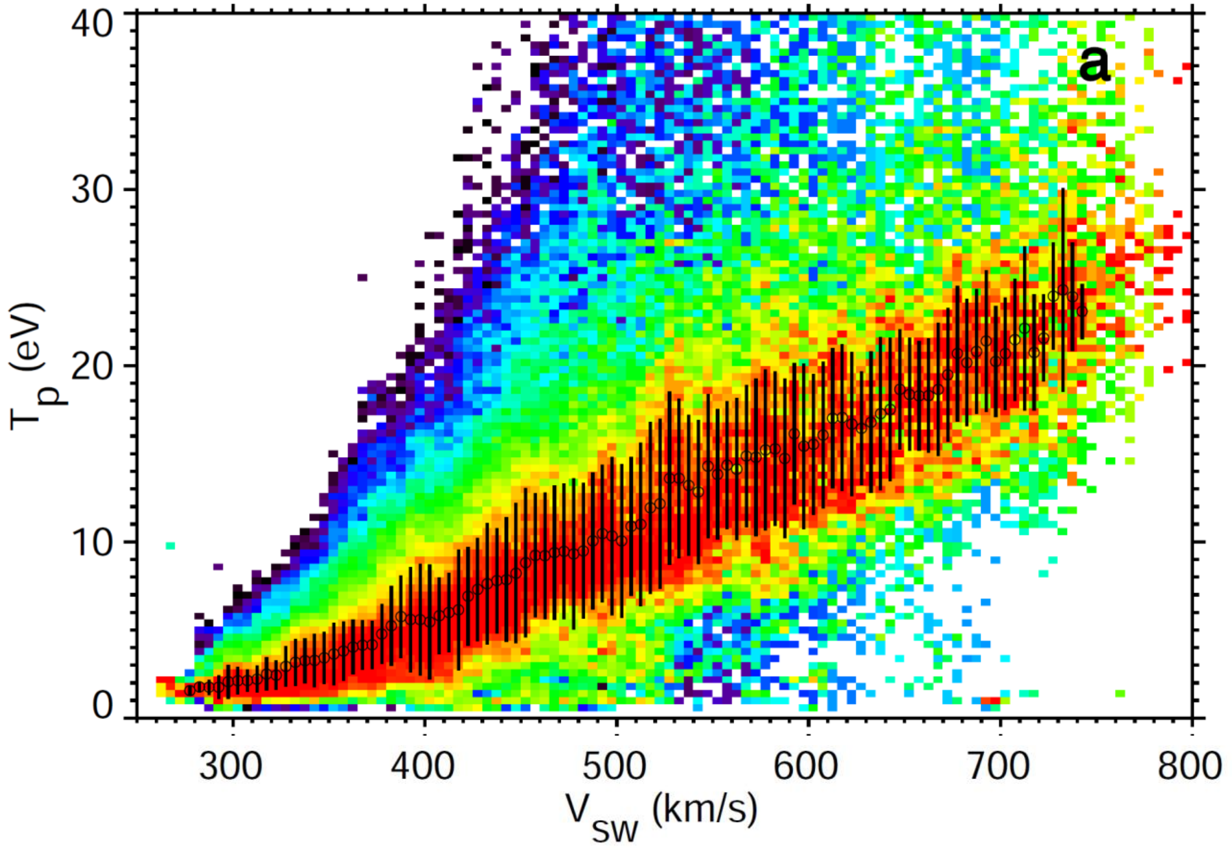
1998 ~ 2011년 태양 극대/극소기 포함 약 13년

Statistic results for input data @ 1 AU

Speed vs Density



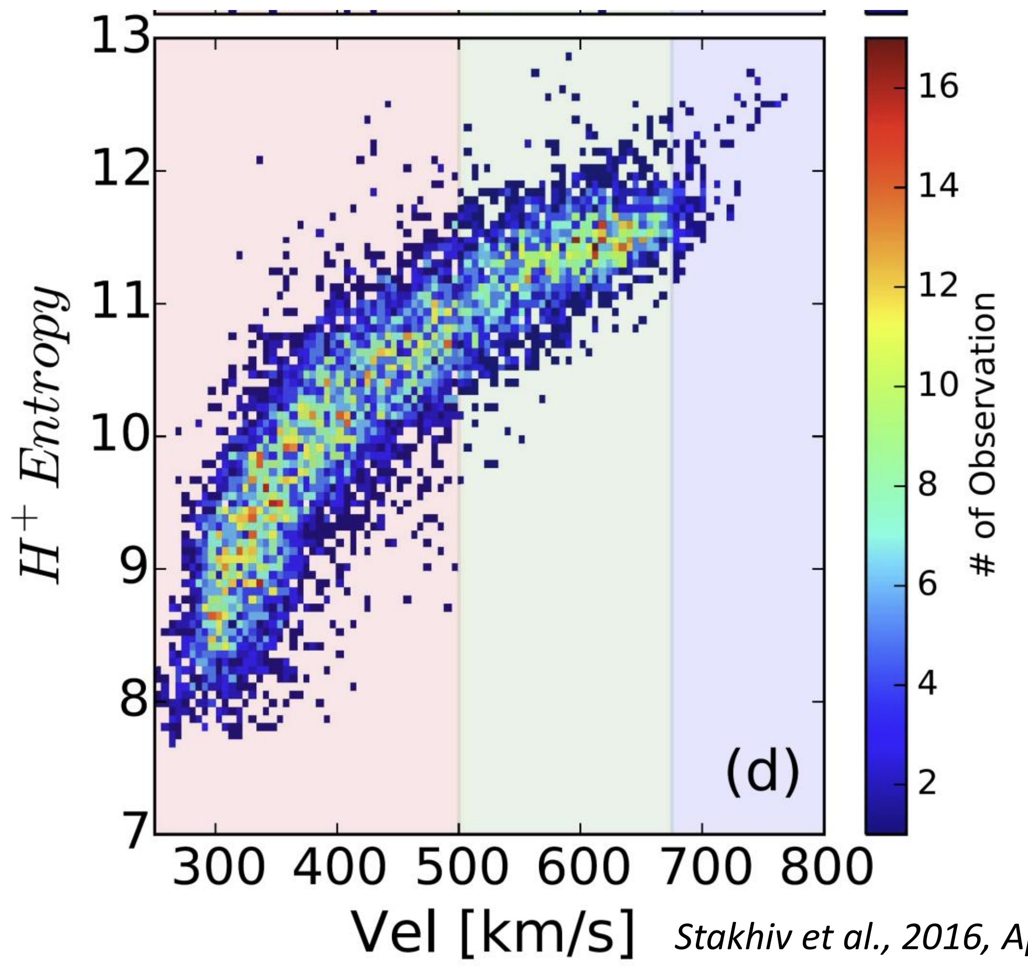
Speed vs Temperature



Salem et al., 2023, A & A

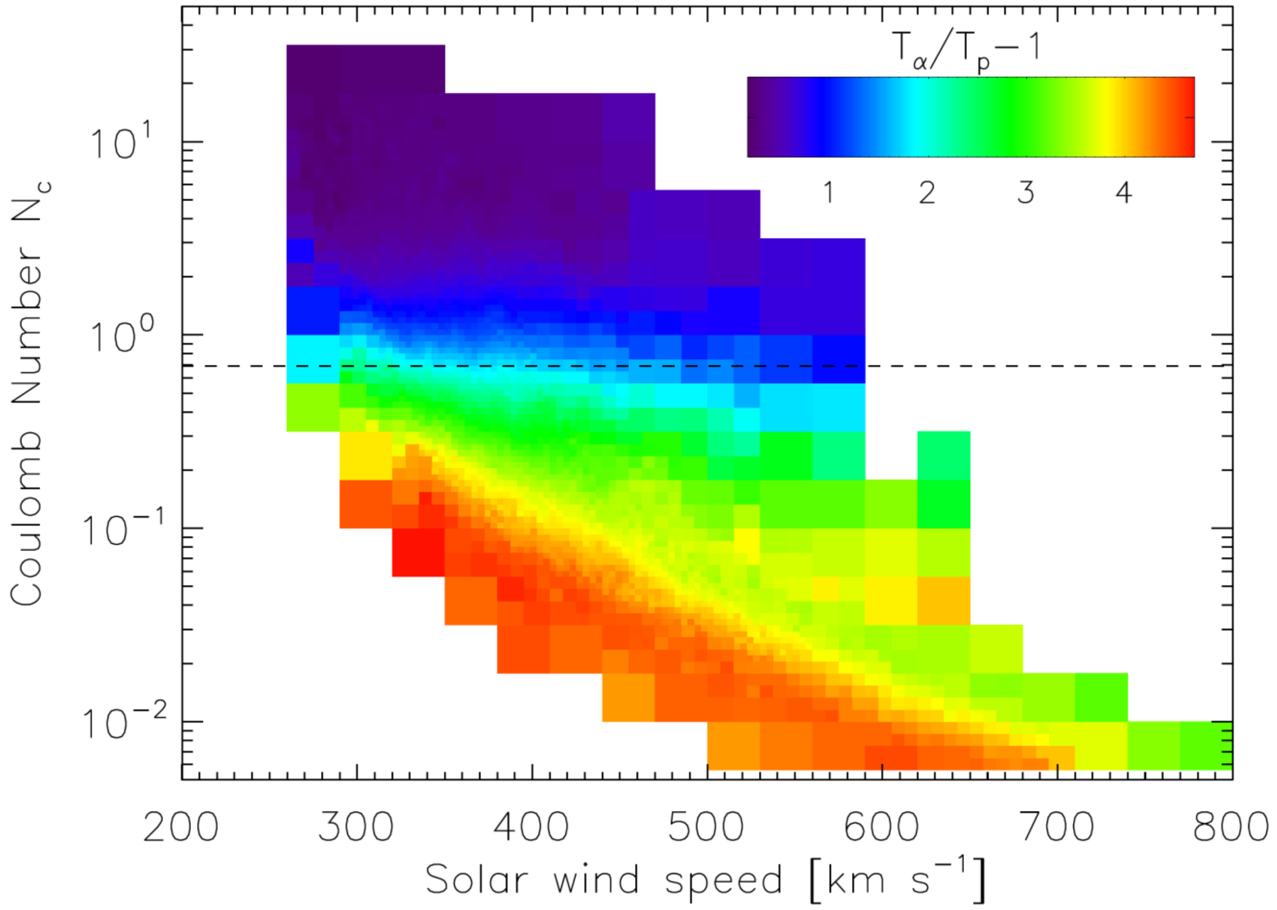
Statistic results for input data @ 1 AU

Speed vs Entropy



Stakhiv et al., 2016, ApJ

Speed vs Coulomb collisions

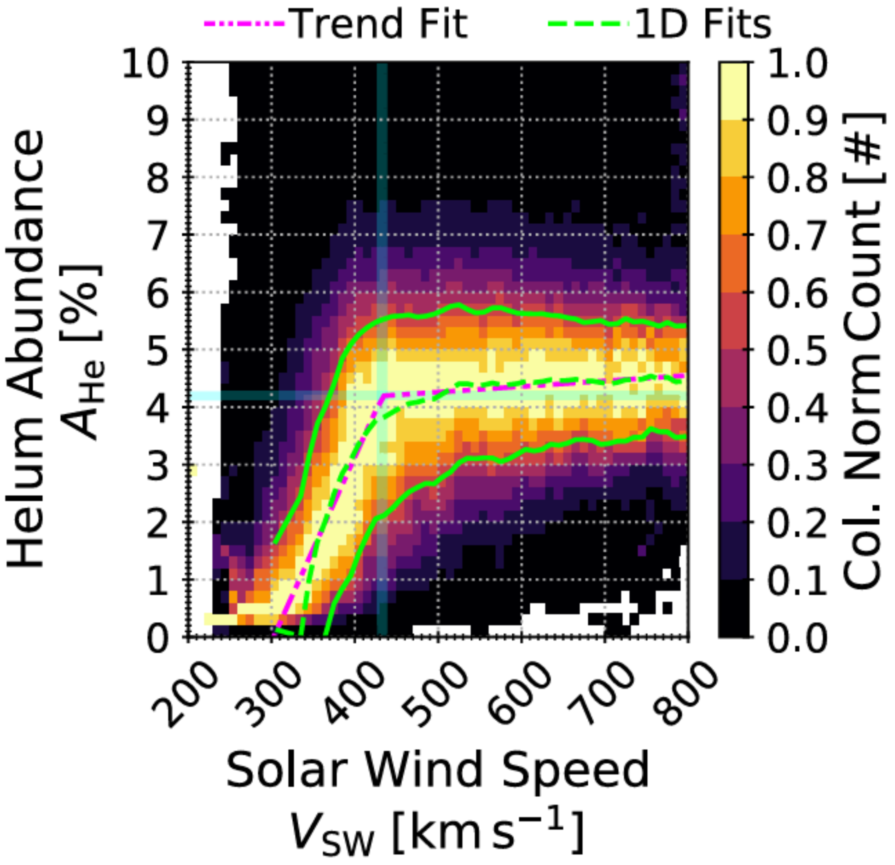


Kasper et al., 2017, ApJ

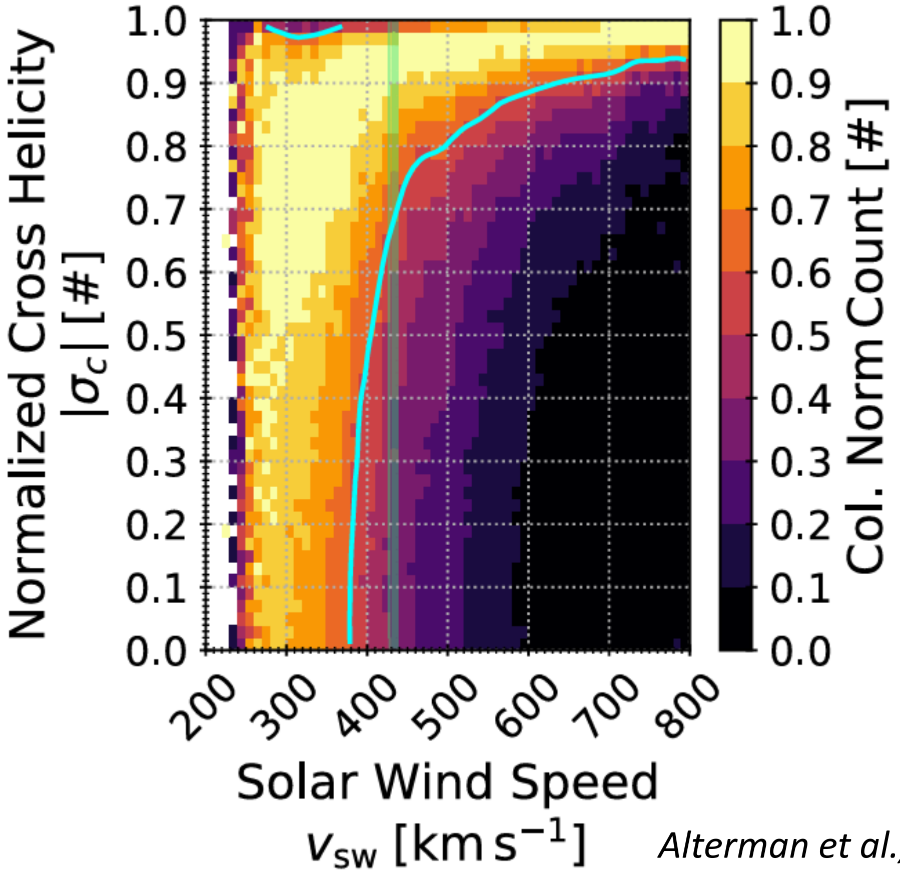
- ✓ *Proton specific entropy:* $\ln\left(\frac{T_p}{n^{1/2}}\right)$
- ✓ *Collisional frequency:* $\frac{n_p}{T_p^{3/2}}$

Statistic results for input data @ 1 AU

Speed vs Helium abundance $\frac{He^{2+}}{H^{+}}$



Speed vs Alfvenicity



Alterman et al., 2025, ApJL

Alfvenicity

- *Alfvenicity is a characteristic property of the observed magnetic fluctuations, namely a high-degree of magnetic field (b)-velocity (v) correlations,*

−1 < Alfvénicity < 1

$$\sigma_c = 2\langle \delta \boldsymbol{V} \cdot \delta \boldsymbol{V}_{\mathrm{A}} \rangle / \langle \delta \boldsymbol{V}^2 + \delta \boldsymbol{V}_{\mathrm{A}}{}^2 \rangle$$

$$(\text{i.e.,} \quad \delta \boldsymbol{V}_{\mathrm{A}} = (\boldsymbol{B} - \langle \boldsymbol{B} \rangle) / \sqrt{\mu_0 \langle \rho \rangle}).$$

$$\delta \boldsymbol{V} = \boldsymbol{V} - \langle \boldsymbol{V} \rangle$$

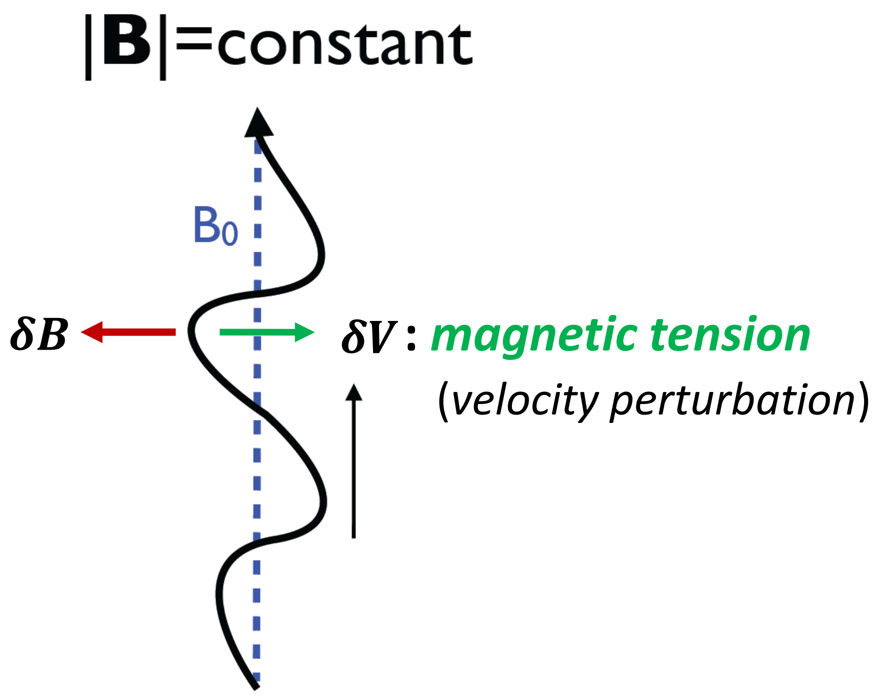
as in large amplitude Alfven waves

"weak compressibility"

$$\boldsymbol{\delta n_p / n_p \ll (\delta v / V_{th})^2,}$$

$V_A = B_0 / (4\pi m_p n_p)^{1/2}$: *Alfven velocity*

$V_{th} = (2 k_B T / m_p)^{1/2}$: *thermal velocity*



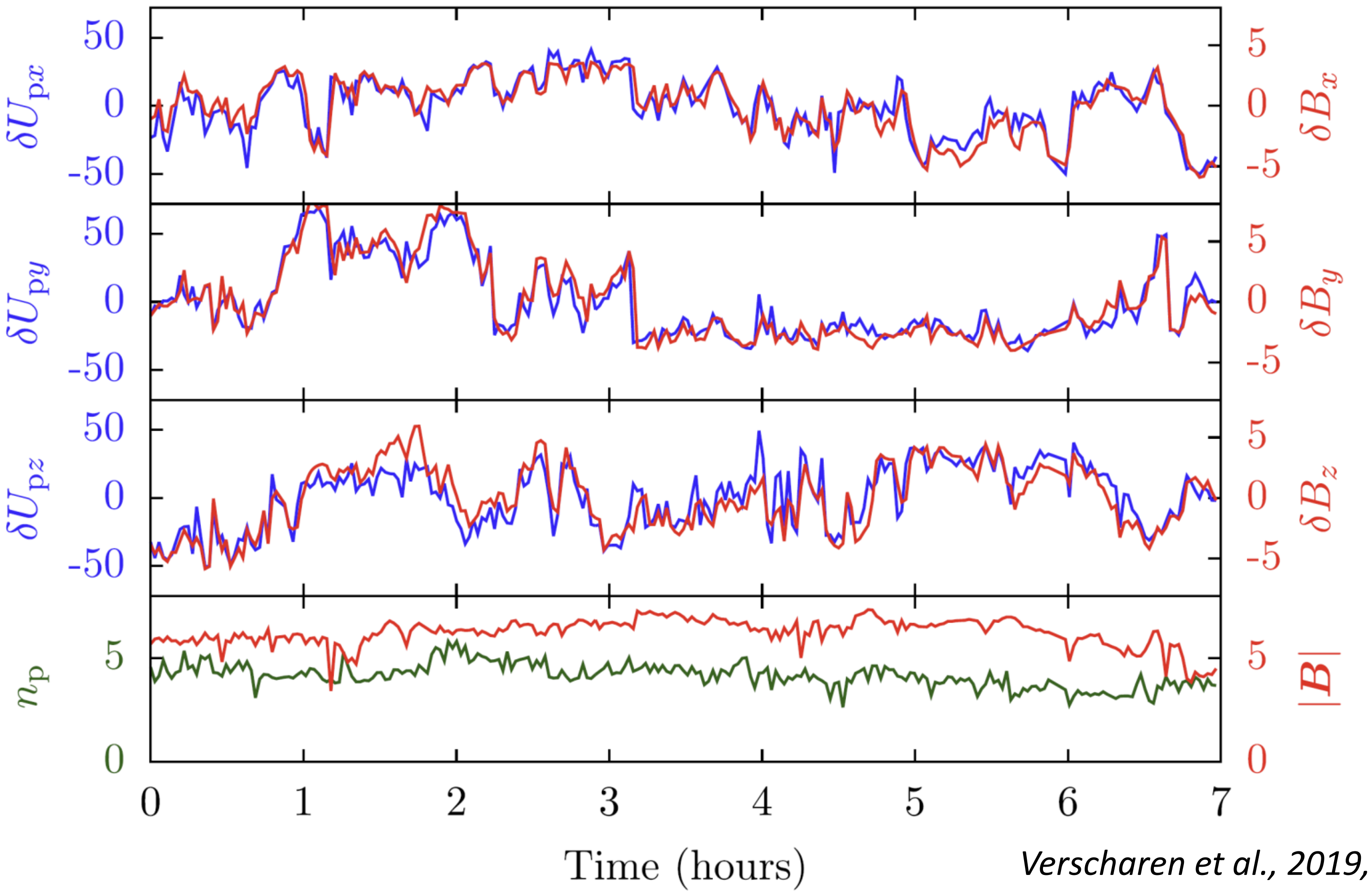
Alfvén waves

Alfvénic correlations

*for **incompressive turbulence***

$$\boxed{\delta \mathbf{v} \simeq \pm \frac{\delta \mathbf{B}}{\sqrt{4\pi\rho}}} \quad sign[-\mathbf{k} \cdot \mathbf{B_0}]$$

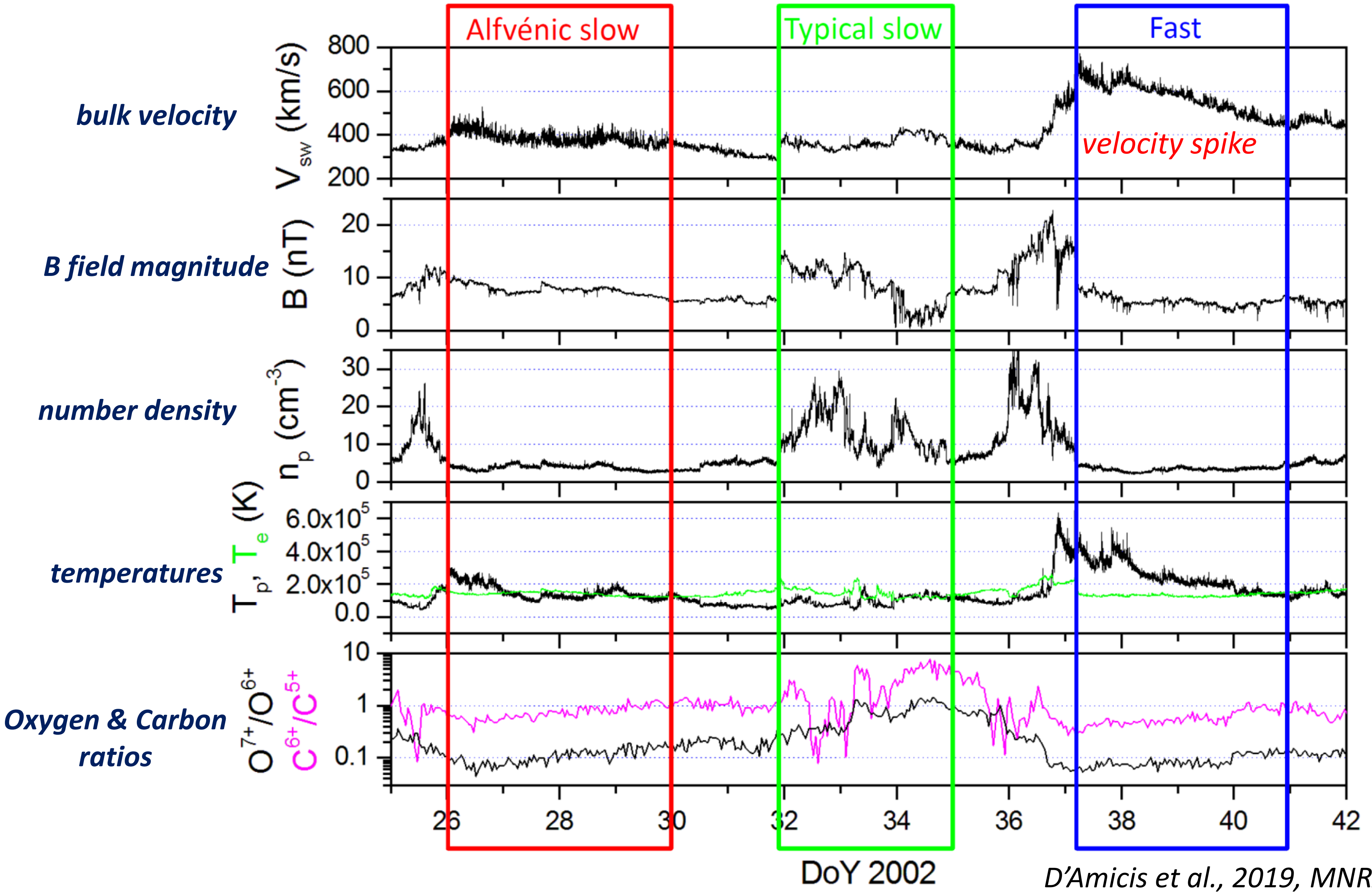
Alfvenic correlation in the solar wind



Verscharen et al., 2019, LRSP

Typical example of three types solar wind

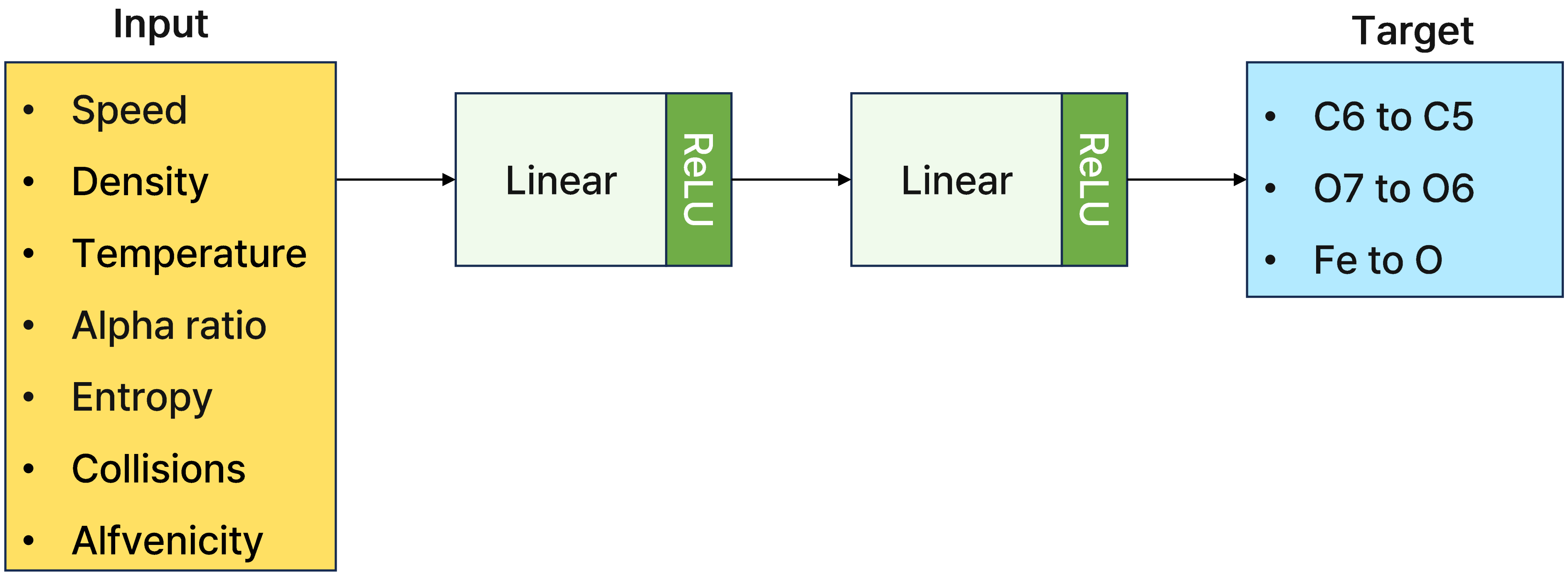
WIND spacecraft measurement at 1 AU



D'Amicis et al., 2019, MNRAS

Deep learning model description

Model Simple MLP



Loss function weighted MSE

$$weight = 1 + (\alpha \ * target)$$
$$loss = weight\ (target - prediction)^2$$

In this experiment, we set α to 3.

Optimizer Adam with learning rate 0.0005

Experiment	Correlation Coefficient			Input parameters
	C^{6+}/C^{5+}	O^{7+}/O^{6+}	Fe/O	
Ex1	0.43	0.67	0.54	Alfvenicity 절대값 없이
Ex2	**0.51**	0.65	0.52	Vp, Alpha_ratio, entropy, **|Alfvenicity|**, Np, **Tp**, coulomb num
Ex3	**0.25**	0.67	0.55	Vp, Alpha_ratio, entropy
Ex4	0.33	**0.69**	**0.55**	Alfvenicity 제외
Ex5	0.50	**0.59**	**0.44**	Vp 제외
Ex6	0.41	0.67	0.54	Alpha ratio 제외
Ex7	0.45	0.62	0.49	Entropy 제외
Ex8	0.38	0.64	0.53	Np 제외
Ex9	0.33	0.64	0.55	Tp 제외
Ex10	0.38	0.66	0.50	Coulomb_num 제외
Ex11	0.41	0.59	0.46	Vp, Alpha_ratio, |Alfventicity|, Np, Tp

Table 1. Results of solar wind ion prediction experiments with varying input parameters.

- ✓ 전반적으로 O^{7+}/O^{6+} 결과의 성능이 좋음.
- ✓ 물리량 계산값을 추가한 것이 모델 학습 결과가 더 좋음.

Deep learning results

Experiment 2 - Vp, Alpha_ratio, entropy, |Alfvenicity|, Np, Tp, coulomb num

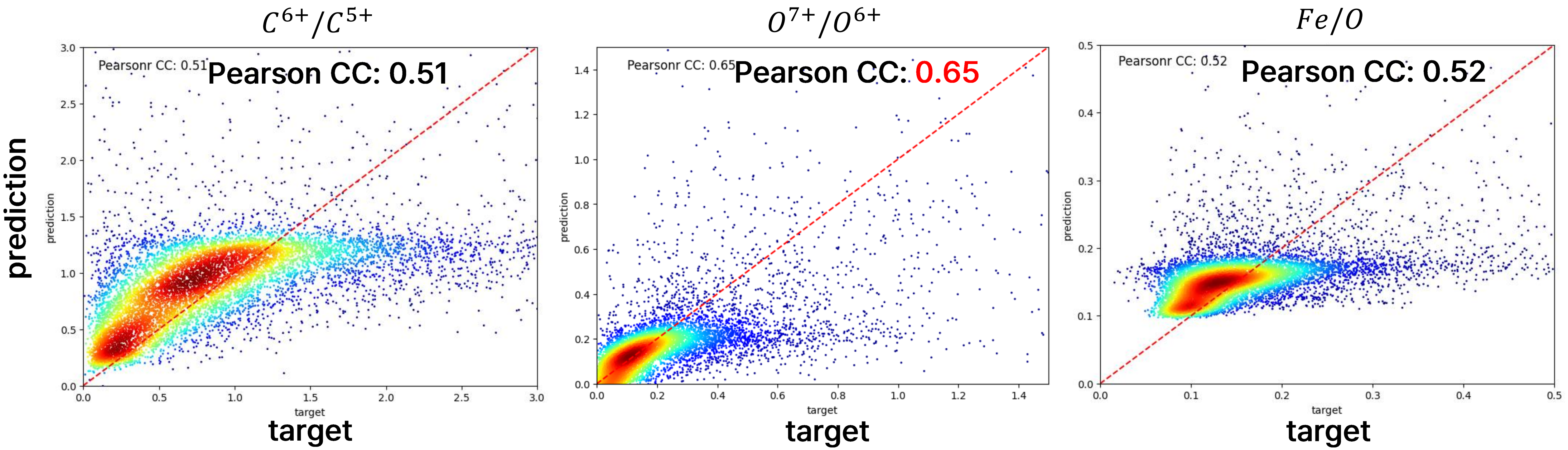
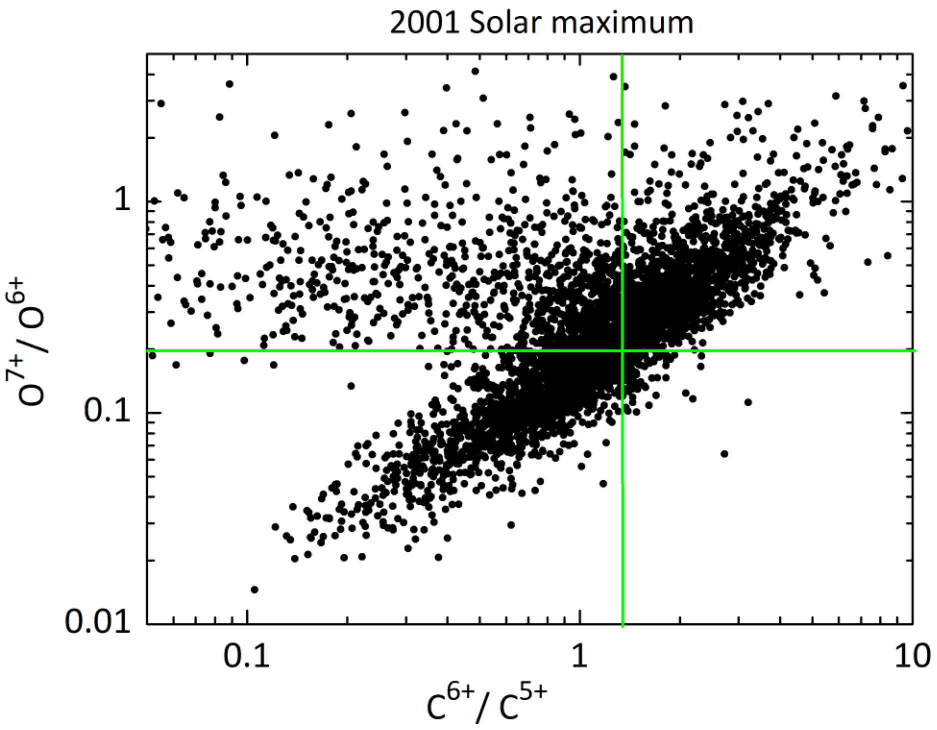
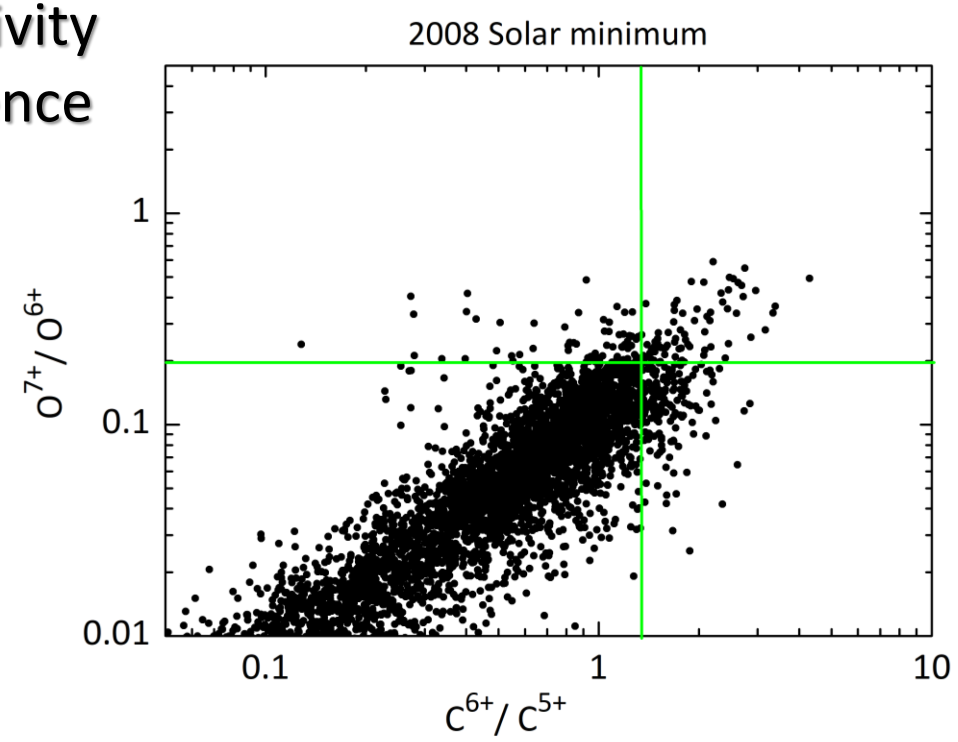
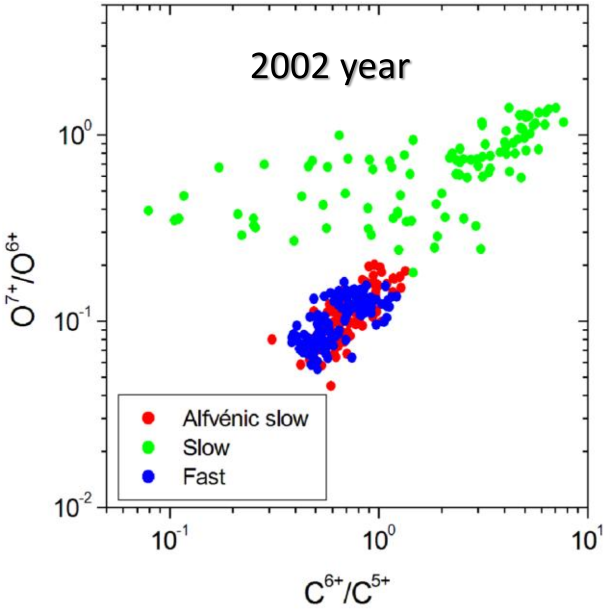


Figure 1. Scatter density plots of predicted from Experiment 2 in Table 1 versus observed ion composition ratios for three species: C^{6+}/C^{5+} (left), O^{7+}/O^{6+} (middle), and Fe/O (right). The red dashed line represents one-to-one line, and the Pearson correlation coefficient (CC) is shown in each panel.

Solar activity dependence







- ***This indicates plasma coming from different source regions***
  : a coronal-hole-associated solar wind - relatively cold source
  : a streamer-associated slow wind - relatively hot source

D'Amicis + 2019

ElasticNet: linear regression

일반적인 선형 모형에 가중치 w를 구할 때 목적 함수에 규제항(regularization) 추가

$$\hat{y} = b + \sum_{j=1}^{d} x_j w_j$$

b: 절편
x_j: 입력 feature 개수 (우리의 경우에는 시간 7 x 변수 7 = 49개)
w_j: 각 feature에 곱해지는 회귀 계수

Experiment				Input parameters
	C^6/C^5	O^7/O^6	Fe/O	
Linear Regression	0.36	0.59	0.41	Vp, Alpha_ratio, entropy, **|Alfvenicity|**, Np, **Tp**, coulomb num

Table 2. Results of solar wind ion prediction using linear regression with same input parameters as Ex2 in Table 1.

- ✓ **Linear regression** 결과는 딥러닝 결과보다 더 낮은 **CC** 를 보여줌
- ✓ 딥러닝 모델이 **linear regression**이 포착하지 못한 **non-linearity**를 학습했을 것.

Linear regression results

- ✓ **Deep learning results**

Pearson CC: 0.51 **Pearson CC: 0.65** **Pearson CC: 0.52**

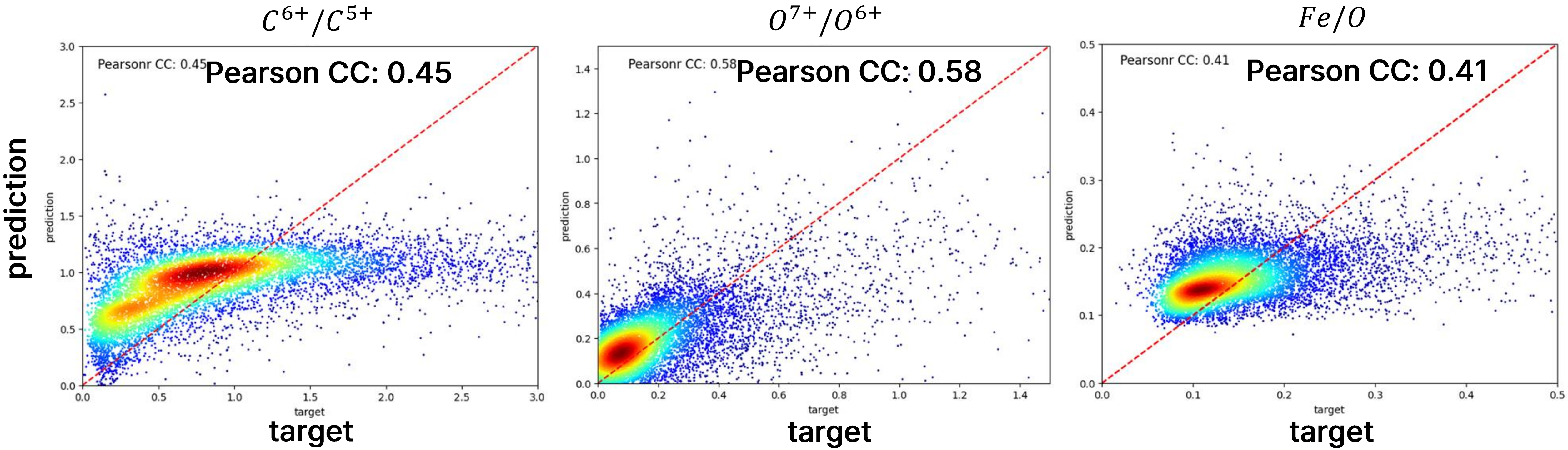
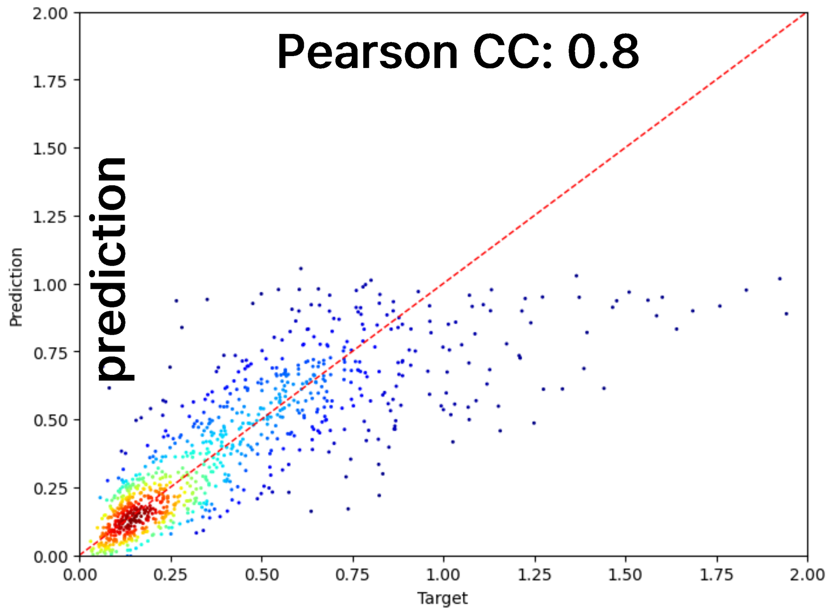


Figure 2. Scatter density plots of predicted from linear regression model versus observed ion composition ratios for three species: C^{6+}/C^{5+} (left), O^{7+}/O^{6+} (middle), and Fe/O (right) of Experiment 2 in Table 1. The red dashed line represents one-to-one line, and the Pearson correlation coefficient (CC) is shown in each panel.

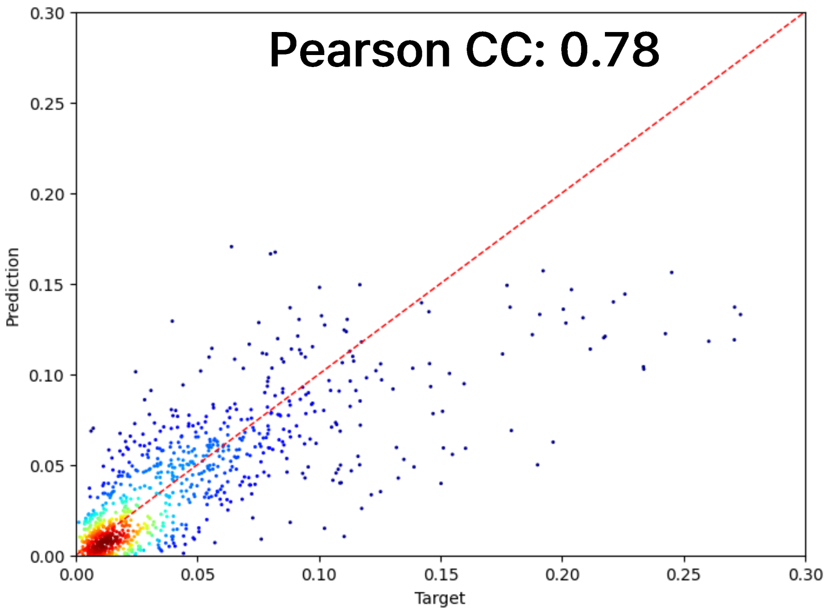
2007-2008
태양활동 극소기

C^{6+}/C^{5+}



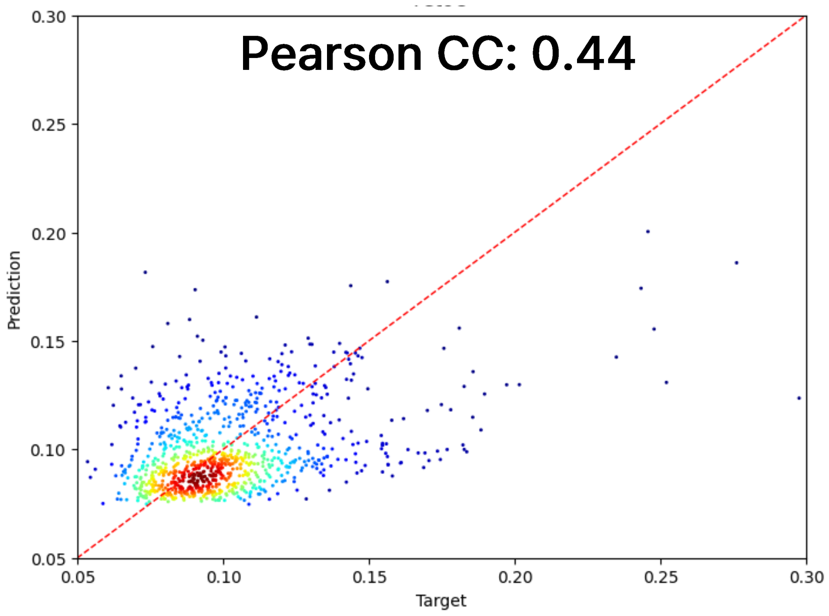
target

O^{7+}/O^{6+}



target

Fe/O



target

Thank You

KASI 한국천문연구원
Korea Astronomy & Space Science Institute